Exercises: Convex Optimization Problems – Answer Key

Mathematical Foundations for Machine Learning

Terminology and Conditions for Convex Optimization

Exercise 1.1 [Easy]
Consider the optimization problem

$$\min_{x\in\mathbb{R}^2} \; x_1^2 + x_2^2 \quad \text{s.t.} \quad x_1 + x_2 \le 3, \quad x_1 \ge 0, \quad x_2 \ge 0.$$

(a) Identify the objective function, the inequality constraint functions, and the feasible set.

(b) Is this a convex optimization problem in standard form? Verify each of the three requirements (convex objective, convex inequality constraints, linear equality constraints).

(c) Is the feasible set convex? Justify your answer.

Solution

(a) The objective function is $f(x) = x_1^2 + x_2^2$. The inequality constraints are $g_1(x) = x_1 + x_2 - 3 \le 0$, $g_2(x) = -x_1 \le 0$, $g_3(x) = -x_2 \le 0$. The feasible set is $\{x \in \mathbb{R}^2 : x_1 + x_2 \le 3,\ x_1 \ge 0,\ x_2 \ge 0\}$, which is a triangle with vertices $(0,0)$, $(3,0)$, $(0,3)$.
(b) Yes. The objective $f(x) = x_1^2 + x_2^2$ is convex (its Hessian is $2I \succ 0$). All constraint functions g_1, g_2, g_3 are affine, hence convex. There are no equality constraints. All three requirements are met.
(c) Yes. The feasible set is the intersection of three half-spaces. Each half-space is convex, and the intersection of convex sets is convex.

Exercise 1.2 [Easy]
For each of the following optimization problems, determine whether it is a convex optimization problem in standard form. If not, state which requirement is violated.

(a) $\min_{x\in\mathbb{R}} \; e^{-x} \quad \text{s.t.} \quad x^2 \le 1$
(b) $\min_{x\in\mathbb{R}^2} \; x_1^2 + x_2^2 \quad \text{s.t.} \quad x_1^2 + x_2^2 = 1$
(c) $\min_{x\in\mathbb{R}^2} \; x_1 + x_2 \quad \text{s.t.} \quad x_1^2 + x_2^2 \le 4, \quad 2x_1 - x_2 = 1$

Solution

(a) Yes, it is convex. The objective e^{-x} is convex (second derivative $e^{-x} > 0$). The constraint $g(x) = x^2 - 1 \le 0$ has a convex constraint function ($g''(x) = 2 > 0$). No equality constraints.
(b) No. The equality constraint $x_1^2 + x_2^2 = 1$ is not linear (it is a nonlinear equality constraint). In standard form, equality constraints must be affine.
(c) Yes. The objective is affine (hence convex). The inequality constraint function $g(x) = x_1^2 + x_2^2 - 4$ is convex. The equality constraint $2x_1 - x_2 = 1$ is linear.

Exercise 1.3 [Medium]
Consider the problem

$$\min_{x\in\mathbb{R}^n} \; \|x\|_2 \quad \text{s.t.} \quad Ax = b,$$

where $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$.

(a) This problem is not in standard convex form. Why not? (Hint: consider differentiability at the origin.)

(b) Show that this problem is equivalent to the convex optimization problem

$$\min_{x\in\mathbb{R}^n} \|x\|_2^2 \quad \text{s.t.} \quad Ax = b.$$

That is, both problems have the same set of optimal solutions.

(c) Verify that this reformulated problem is a convex optimization problem in standard form. Identify the problem class (LP, QP, QCQP, or SOCP).

Solution

(a) The objective $f(x) = \|x\|_2$ is convex but not differentiable at $x = 0$. More importantly, the issue is that while $\|x\|_2$ is indeed convex and the problem has convex feasible set and convex objective, the problem is technically in standard form. However, the practical concern is differentiability. The deeper reason to reformulate: $\|x\|_2$ is not *strictly* convex (it is constant on rays), which makes analysis harder.
Note: Actually $\|x\|_2$ *is* convex, so the problem is in standard convex form. The point of the exercise is that the equivalent squared formulation is *strictly* convex and smooth, making it easier to solve.
(b) Since $t \mapsto t^2$ is strictly increasing on $[0, \infty)$ and $\|x\|_2 \geq 0$ for all x, we have

$$\|x^*\|_2 \leq \|x\|_2 \quad \Longleftrightarrow \quad \|x^*\|_2^2 \leq \|x\|_2^2$$

for all feasible x. Therefore x^* minimizes $\|x\|_2$ over the feasible set if and only if it minimizes $\|x\|_2^2$ over the same set.
(c) The objective $f(x) = \|x\|_2^2 = x^T I x$ is a convex quadratic (Hessian $= 2I \succ 0$). The constraint $Ax = b$ is linear. Hence this is a convex optimization problem in standard form. It is a **QP** (quadratic program): $\min \frac{1}{2}x^T(2I)x$ subject to $Ax = b$, with $P = 2I \in \mathbb{S}_+^n$.

Exercise 1.4 [Medium]
Consider the problem of minimizing the maximum component of a vector subject to linear constraints:

$$\min_{x\in\mathbb{R}^n} \max_{i=1,\dots,n} x_i \quad \text{s.t.} \quad Ax = b.$$

(a) Is this a convex optimization problem in standard form? Justify.

(b) Reformulate it as an equivalent **linear program** by introducing an auxiliary variable $t \in \mathbb{R}$. Write the LP in standard form.

(c) What general principle does this illustrate about equivalent convex problems?

Solution

(a) The objective $f(x) = \max_i x_i$ is convex (it is the pointwise maximum of affine functions). The constraint is linear. So yes, this is a convex optimization problem. However, it is not an LP in its current form because the objective is piecewise linear, not affine.
(b) Introduce $t \in \mathbb{R}$ and reformulate using the epigraph:

$$\min_{x\in\mathbb{R}^n, t\in\mathbb{R}} \quad t \quad \text{s.t.} \quad x_i \le t,\ i = 1, \dots, n, \quad Ax = b.$$

This is an LP: the objective is linear in (x, t), the inequality constraints $x_i - t \le 0$ are affine (hence convex), and the equality constraint is linear.
(c) This illustrates the **epigraph reformulation**: any convex optimization problem can be reformulated by minimizing over the epigraph. In particular, minimizing a piecewise-linear convex function can always be transformed into an LP. This is one of the key transformations that preserves convexity and often simplifies the problem class.

Local Optimality Implies Global Optimality

Exercise 2.1 [Easy]
Let $\Omega \subseteq \mathbb{R}^n$ be a convex set and $f : \Omega \to \mathbb{R}$ be a convex function. Suppose $x^* \in \Omega$ is a local minimizer, i.e., there exists $R > 0$ such that $f(x^*) \le f(z)$ for all feasible z with $\|z - x^*\| \le R$.

Prove that x^* is a global minimizer: $f(x^*) \le f(y)$ for all $y \in \Omega$.

Solution

Suppose for contradiction that x^* is a local minimizer but not a global minimizer. Then there exists a feasible $y \in \Omega$ such that $f(y) < f(x^*)$.
Since Ω is convex, for any $\lambda \in (0, 1)$ the point $z_\lambda = \lambda y + (1 - \lambda)x^*$ is feasible.
By convexity of f:

$$f(z_\lambda) = f(\lambda y + (1 - \lambda)x^*) \le \lambda f(y) + (1 - \lambda)f(x^*) < \lambda f(x^*) + (1 - \lambda)f(x^*) = f(x^*).$$

Now, $\|z_\lambda - x^*\| = \lambda\|y - x^*\|$. By choosing λ small enough, we can ensure $\|z_\lambda - x^*\| < R$. But then z_λ is in the local neighborhood of x^* and $f(z_\lambda) < f(x^*)$, which contradicts the local optimality of x^*.
Therefore x^* must be a global minimizer. $\square$

Exercise 2.2 [Medium]
(a) Let $f : \mathbb{R}^n \to \mathbb{R}$ be strictly convex. Prove that if a global minimizer x^* exists, then it is unique.

(b) Give an example of a strictly convex function $f : \mathbb{R} \to \mathbb{R}$ that does *not* attain its global minimum. What additional condition guarantees existence?

(c) Show that if f is strongly convex with parameter $m > 0$, then f is coercive, i.e., $f(x) \to \infty$ as $\|x\| \to \infty$. Conclude that a strongly convex, lower semi-continuous function on a closed convex domain always has a unique global minimizer.

Solution

(a) Suppose x^* and y^* are both global minimizers with $x^* \neq y^*$, so $f(x^*) = f(y^*) = f^*$. Since the feasible set is convex, $z = \frac{x^*+y^*}{2}$ is feasible. By strict convexity:

$$f(z) = f\left(\frac{x^* + y^*}{2}\right) < \frac{1}{2}f(x^*) + \frac{1}{2}f(y^*) = f^*,$$

which contradicts the global optimality of x^*. Hence $x^* = y^*$. $\square$

(b) The function $f(x) = e^{-x}$ is strictly convex ($f''(x) = e^{-x} > 0$) but $\inf_{x\in\mathbb{R}} e^{-x} = 0$, which is never attained. The additional condition is **coercivity**: $f(x) \to \infty$ as $\|x\| \to \infty$ (equivalently, sublevel sets are bounded). Together with lower semi-continuity and a closed domain, this guarantees existence.

(c) By strong convexity with parameter m, for any x, y:

$$f(y) \geq f(x) + \nabla f(x)^T(y - x) + \frac{m}{2}\|y - x\|^2.$$

Fix $x = x_0$. As $\|y\| \to \infty$, the term $\frac{m}{2}\|y - x_0\|^2$ dominates the linear term, so $f(y) \to \infty$. Thus f is coercive.

Coercivity implies that for any level α, the sublevel set $\{x : f(x) \leq \alpha\}$ is bounded. Combined with lower semi-continuity and closedness of the domain, the sublevel sets are compact. By the extreme value theorem (Lemma 16), f attains its minimum. By strict convexity (which follows from strong convexity), the minimizer is unique. $\square$

Exercise 2.3 [Hard]

Let $f : \Omega \to \mathbb{R}$ be convex and continuously differentiable on a convex set Ω, and denote by $X \subseteq \Omega$ the feasible set.

(a) Prove that $x^* \in X$ is optimal if and only if

$$\langle \nabla f(x^*), y - x^* \rangle \geq 0, \quad \forall\, y \in X.$$

(Hint: use the first-order characterization of convexity for one direction, and a directional argument for the other.)

(b) Suppose now $X = \{x : Ax = b\}$. Show that the optimality condition reduces to: there exists $\nu \in \mathbb{R}^s$ such that $\nabla f(x^*) + A^T\nu = 0$.

(Hint: every feasible direction is of the form $y - x^* = v$ where $v \in \ker(A)$.)

Solution

(a) ($\Leftarrow$): Suppose $x^* \in X$ and $\langle \nabla f(x^*), y - x^* \rangle \geq 0$ for all $y \in X$. By the first-order characterization of convexity:

$$f(y) \geq f(x^*) + \langle \nabla f(x^*), y - x^* \rangle \geq f(x^*)$$

for all feasible y. Hence x^* is globally optimal.
($\Rightarrow$): Suppose x^* is optimal but there exists $y \in X$ with $\langle \nabla f(x^*), y - x^* \rangle < 0$. Consider $z(t) = x^* + t(y - x^*)$ for $t \in [0, 1]$. Since X is convex, $z(t) \in X$ for all $t \in [0, 1]$. Then:

$$\frac{d}{dt} f(z(t)) \bigg|_{t=0} = \langle \nabla f(x^*), y - x^* \rangle < 0.$$

So for small enough $t > 0$, $f(z(t)) < f(x^*)$, contradicting the optimality of x^*. $\square$
(b) With $X = \{x : Ax = b\}$, every feasible point y satisfies $Ay = b = Ax^*$, so $A(y - x^*) = 0$, meaning $y - x^* \in \ker(A)$. The condition becomes:

$$\langle \nabla f(x^*), v \rangle \geq 0, \quad \forall\, v \in \ker(A).$$

Since $\ker(A)$ is a subspace, if $v \in \ker(A)$ then $-v \in \ker(A)$, so we also need $\langle \nabla f(x^*), -v \rangle \geq 0$. Together: $\langle \nabla f(x^*), v \rangle = 0$ for all $v \in \ker(A)$.
This means $\nabla f(x^*) \in \ker(A)^\perp = \operatorname{ran}(A^T)$. Therefore there exists $\nu \in \mathbb{R}^s$ such that $\nabla f(x^*) = -A^T \nu$, i.e., $\nabla f(x^*) + A^T \nu = 0$. $\square$

Linear and Quadratic Programs

Exercise 3.1 [Easy]
Classify each of the following problems as an LP, QP, QCQP, SOCP, or none of these. Justify briefly.

(a) $\min_{x \in \mathbb{R}^3}\ 2x_1 - x_2 + 4x_3 \quad \text{s.t.} \quad x_1 + x_2 + x_3 = 1,\ x_i \geq 0$
(b) $\min_{x \in \mathbb{R}^2}\ x_1^2 + 3x_2^2 + 2x_1x_2 \quad \text{s.t.} \quad x_1 + x_2 \leq 5$
(c) $\min_{x \in \mathbb{R}^2}\ x_1 + x_2 \quad \text{s.t.} \quad x_1^2 + x_2^2 \leq 4,\ x_1^2 - x_2 \leq 0$

Solution

(a) LP. The objective $2x_1 - x_2 + 4x_3$ is affine. The equality constraint is linear. The non-negativity constraints $-x_i \leq 0$ are linear. This is an LP in standard form.
(b) QP. The objective is quadratic: $f(x) = x^T P x$ where $P = \begin{pmatrix} 1 & 1 \\ 1 & 3 \end{pmatrix}$. We verify $P \succeq 0$: eigenvalues are $2 \pm \sqrt{2} > 0$, so $P \succ 0$. The constraint is linear (polyhedral). Quadratic objective over a polyhedron = QP.
(c) QCQP. The objective is linear (hence quadratic with $P = 0$). The constraint $x_1^2 + x_2^2 \leq 4$ is a quadratic constraint with $P_1 = I \succeq 0$ (convex). The constraint $x_1^2 - x_2 \leq 0$ can be written as $\frac{1}{2} x^T P_2 x + q_2^T x \leq 0$ with $P_2 = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} \succeq 0$ and $q_2 = (0, -1)^T$. Both quadratic constraint functions are convex, so this is a QCQP.

Exercise 3.2 [Medium]
A company produces n products. The profit per unit of product i is $c_i > 0$. Production

of product i requires a_{ji} units of resource j, and there are b_j units of resource j available ($j = 1, \ldots, m$). Let x_i be the quantity produced of product i.

(a) Formulate the profit maximization problem as a linear program.

(b) Convert this to a minimization LP in standard form as defined in the lecture (with equality constraints and non-negativity).

(c) Now suppose the profit per unit is uncertain: c_i is a random variable with mean $\bar{c}_i$ and the cost vector c has covariance $\Sigma \succ 0$. The company wants to maximize the risk-adjusted profit $\mathbb{E}[c^T x] - \gamma \operatorname{Var}(c^T x)$ for some $\gamma > 0$. Formulate this as a QP.

Solution

(a) Maximize profit subject to resource and non-negativity constraints:

$$\max_{x \in \mathbb{R}^n} \; c^T x \quad \text{s.t.} \quad Ax \leq b, \quad x \geq 0,$$

where $A \in \mathbb{R}^{m \times n}$ with entries a_{ji} and $b = (b_1, \ldots, b_m)^T$.

(b) Convert to minimization: $\min -c^T x$. Introduce slack variables $s \in \mathbb{R}^m$ with $s \geq 0$ to convert inequalities to equalities:

$$\min_{x,s} \; -c^T x \quad \text{s.t.} \quad Ax + s = b, \quad x \geq 0, \quad s \geq 0.$$

Letting $\tilde{x} = (x, s)^T$, $\tilde{c} = (-c, 0)^T$, $\tilde{A} = (A \mid I)$, this is $\min \tilde{c}^T \tilde{x}$ s.t. $\tilde{A}\tilde{x} = b$, $\tilde{x} \geq 0$.

(c) We have $\mathbb{E}[c^T x] = \bar{c}^T x$ and $\operatorname{Var}(c^T x) = x^T \Sigma x$. The risk-adjusted objective is:

$$\max_{x} \; \bar{c}^T x - \gamma x^T \Sigma x \quad \text{s.t.} \quad Ax \leq b, \; x \geq 0.$$

Equivalently, as a minimization QP:

$$\min_{x} \; \gamma x^T \Sigma x - \bar{c}^T x \quad \text{s.t.} \quad Ax \leq b, \; x \geq 0.$$

Since $\Sigma \succ 0$ and $\gamma > 0$, the matrix $P = 2\gamma\Sigma \in \mathbb{S}^n_+$, confirming this is a valid QP.

Exercise 3.3 [Medium]

Consider Ridge Regression: given data $X \in \mathbb{R}^{N \times D}$, targets $y \in \mathbb{R}^N$, and regularization parameter $\lambda > 0$:

$$\min_{\theta \in \mathbb{R}^D} \; \|X\theta - y\|_2^2 + \lambda \|\theta\|_2^2.$$

(a) Show this is a QP by identifying the matrix P, the vector q, and the constant c in the standard form $\min \frac{1}{2}\theta^T P \theta + q^T \theta + c$.

(b) Prove that $P \succ 0$ for any $\lambda > 0$ and any X, and conclude that the problem is strictly convex with a unique minimizer.

(c) Derive the closed-form solution by setting the gradient to zero.

Solution

(a) Expand the objective:

$$\|X\theta - y\|^2 + \lambda\|\theta\|^2 = \theta^T X^T X\theta - 2y^T X\theta + y^T y + \lambda\theta^T\theta = \theta^T(X^T X + \lambda I)\theta - 2y^T X\theta + \|y\|^2.$$

In standard form: $P = 2(X^T X + \lambda I)$, $q = -2X^T y$, $c = \|y\|^2$. This is an unconstrained QP.
(b) For any $v \neq 0$:

$$v^T(X^T X + \lambda I)v = \|Xv\|^2 + \lambda\|v\|^2 \geq \lambda\|v\|^2 > 0.$$

So $X^T X + \lambda I \succ 0$, hence $P = 2(X^T X + \lambda I) \succ 0$. A QP with $P \succ 0$ is strictly convex, so by Theorem 68, the global minimizer exists and is unique.
(c) Set $\nabla_\theta f = 0$:

$$2(X^T X + \lambda I)\theta - 2X^T y = 0 \implies (X^T X + \lambda I)\theta = X^T y.$$

Since $X^T X + \lambda I \succ 0$ (hence invertible):

$$\hat{\theta} = (X^T X + \lambda I)^{-1} X^T y.$$

Exercise 3.4 [Hard]
Consider the Chebyshev center problem: find the center x_c and largest radius R of a Euclidean ball $B(x_c, R) = \{x_c + u : \|u\|_2 \leq R\}$ contained in a polyhedron $P = \{x \in \mathbb{R}^n : \langle a_i, x\rangle \leq b_i,\ i = 1, \ldots, r\}$.

(a) Show that the constraint $B(x_c, R) \subseteq \{x : \langle a_i, x\rangle \leq b_i\}$ can be written as

$$\langle a_i, x_c\rangle + R\|a_i\|_2 \leq b_i.$$

(Hint: the ball lies in the half-space iff $\sup\{\langle a_i, x_c + u\rangle : \|u\| \leq R\} \leq b_i$.)

(b) Formulate the full optimization problem and show it is a linear program in the variables (x_c, R).

(c) Explain why this is a convex optimization problem even though we are *maximizing* R.

Solution

(a) The ball $B(x_c, R)$ lies in the half-space $\{x : \langle a_i, x\rangle \le b_i\}$ if and only if:

$$\sup_{\|u\|\le R} \langle a_i, x_c + u\rangle \le b_i.$$

We compute the supremum:

$$\sup_{\|u\|\le R} \langle a_i, x_c + u\rangle = \langle a_i, x_c\rangle + \sup_{\|u\|\le R} \langle a_i, u\rangle.$$

By the Cauchy-Schwarz inequality, $\langle a_i, u\rangle \le \|a_i\|\|u\| \le R\|a_i\|$, with equality when $u = R\frac{a_i}{\|a_i\|}$. Therefore:

$$\langle a_i, x_c\rangle + R\|a_i\|_2 \le b_i. \quad \square$$

(b) We want the largest ball, so we maximize R:

$$\max_{x_c\in\mathbb{R}^n, R\ge 0} R \quad \text{s.t.} \quad \langle a_i, x_c\rangle + R\|a_i\|_2 \le b_i,\ i = 1, \dots, r.$$

The objective R is linear in (x_c, R). Each constraint is affine in (x_c, R) (note $\|a_i\|_2$ is a known constant). Hence this is an LP.

(c) Maximizing a linear (hence concave) objective is equivalent to minimizing $-R$, which is convex. The constraints are affine (hence convex). Therefore the problem is a convex optimization problem. In general, $\max f(x)$ with f concave and convex constraints is equivalent to $\min(-f(x))$ with $-f$ convex, which is a standard convex problem.

QCQP and SOCP

Exercise 4.1 [Medium]

Consider the robust linear program: given uncertainty sets $\mathcal{E}_i = \{\bar{a}_i + P_i u : \|u\|_2 \le 1\}$ with $P_i \in \mathbb{S}^n_+$, we want

$$\min_{x\in\mathbb{R}^n} c^T x \quad \text{s.t.} \quad \langle a_i, x\rangle \le b_i, \quad \forall\, a_i \in \mathcal{E}_i,\ i = 1, \dots, r.$$

(a) Show that the robust constraint $\langle a_i, x\rangle \le b_i$ for all $a_i \in \mathcal{E}_i$ is equivalent to

$$\langle \bar{a}_i, x\rangle + \|P_i x\|_2 \le b_i.$$

(b) Is this constraint linear, quadratic, or a second-order cone constraint? What problem class does the full robust LP belong to?

Solution

(a) The robust constraint requires:

$$\sup_{a_i \in \mathcal{E}_i} \langle a_i, x \rangle \leq b_i.$$

Substituting $a_i = \bar{a}_i + P_i u$ with $\|u\|_2 \leq 1$:

$$\sup_{\|u\| \leq 1} \langle \bar{a}_i + P_i u, x \rangle = \langle \bar{a}_i, x \rangle + \sup_{\|u\| \leq 1} \langle u, P_i x \rangle.$$

By Cauchy-Schwarz: $\sup_{\|u\| \leq 1} \langle u, P_i x \rangle = \|P_i x\|_2$, achieved at $u = \frac{P_i x}{\|P_i x\|}$.
Therefore the robust constraint is: $\langle \bar{a}_i, x \rangle + \|P_i x\|_2 \leq b_i$. $\square$
(b) This is a **second-order cone (SOC) constraint**. It has the form $\|A_i x + b_i'\|_2 \leq c_i^T x + d_i$ with $A_i = P_i$, $b_i' = 0$, $c_i = -\bar{a}_i$, $d_i = b_i$ (after rearranging). The objective is linear. Therefore the full problem is a **second-order cone program (SOCP)**.
Note the hierarchy: if $P_i = 0$ for all i, the SOC constraints reduce to linear constraints and the SOCP becomes an LP.

Exercise 4.2 [Hard]
Show the inclusion chain LP $\subset$ QP $\subset$ QCQP $\subset$ SOCP by answering:

(a) Explain why every LP is a QP. What is the matrix P in the quadratic objective?

(b) Explain why every QP is a QCQP.

(c) Show that a convex quadratic constraint $\frac{1}{2} x^T P x + q^T x + c \leq 0$ with $P \in \mathbb{S}_+^n$ can be written as a second-order cone constraint. (Hint: use the Cholesky factorization $P = LL^T$ and complete the square.)

(d) For each inclusion, give a concrete example showing the inclusion is strict (i.e., a problem in the larger class that is *not* in the smaller class).

Solution

(a) An LP has objective $c^T x$, which equals $\frac{1}{2}x^T Px + q^T x$ with $P = 0$ (the zero matrix, which is PSD) and $q = c$. The constraints are linear (polyhedral), which is a special case of polyhedral constraints in a QP.
(b) A QP has a quadratic objective and linear (polyhedral) constraints. Linear constraints $g^T x \leq h$ can be written as quadratic constraints $\frac{1}{2}x^T 0x + g^T x - h \leq 0$ with $P_i = 0 \in \mathbb{S}^n_+$. So every QP is a QCQP.
(c) Let $P = LL^T$ (Cholesky). The constraint $\frac{1}{2}x^T LL^T x + q^T x + c \leq 0$ can be rewritten. Let $y = L^T x$. Then $\frac{1}{2}\|y\|^2 + q^T x + c \leq 0$, i.e., $\|L^T x\|^2 \leq -2q^T x - 2c$. Assuming the right side is non-negative at feasible points, taking square roots:

$$\|L^T x\|_2 \leq \sqrt{-2q^T x - 2c}.$$

More precisely, by completing the square and using the epigraph, one can show this is equivalent to the SOC constraint $\|L^T x\|_2 \leq d^T x + e$ for appropriate d, e derived from q and c, after introducing auxiliary variables if needed. The key insight: a convex quadratic constraint defines an ellipsoid, which is the image of a ball under an affine map, and balls are SOC-representable.
(d) Strict inclusions:

- LP $\subsetneq$ QP: $\min x^2$ subject to $x \leq 1$ is a QP but not an LP (quadratic objective).
- QP $\subsetneq$ QCQP: $\min x_1$ subject to $x_1^2 + x_2^2 \leq 1$ has a quadratic constraint (ellipsoidal feasible set, not polyhedral), so it is a QCQP but not a QP.
- QCQP $\subsetneq$ SOCP: $\min c^T x$ subject to $\|Ax + b\|_2 \leq d^T x + e$ with general A is an SOCP. When A is not square or the constraint cannot be squared to a single quadratic, this goes beyond QCQP.

Mixed Challenging Exercises

Exercise 5.1 [Hard]
Consider the problem of finding the point in a convex polyhedron $P = \{x \in \mathbb{R}^n : Ax \leq b\}$ closest to a given point $p \in \mathbb{R}^n$:

$$\min_{x \in \mathbb{R}^n} \|x - p\|_2^2 \quad \text{s.t.} \quad Ax \leq b.$$

(a) Classify this as an LP, QP, QCQP, or SOCP. Identify P, q, c in standard quadratic form.
(b) Prove that this problem always has a unique solution (state clearly which theorems you use).
(c) Use the optimality criterion for differentiable convex functions: show that x^* is optimal if and only if $\langle x^* - p, y - x^* \rangle \geq 0$ for all $y \in P$. Give a geometric interpretation of this condition.
(d) In the special case $P = \{x : x_1 + x_2 \geq 4,\ 2x_1 + x_2 \geq 5\}$ and $p = (0, 0)^T$, use the KKT conditions to find the closest point to the origin.

Solution

(a) This is a **QP**. Expanding: $\|x-p\|^2 = x^Tx - 2p^Tx + \|p\|^2$. In standard form: $P = 2I \in \mathbb{S}^n_{++}$, $q = -2p$, $c = \|p\|^2$. The constraints are linear.
(b) The objective $f(x) = \|x-p\|^2$ is strongly convex with parameter $m = 2$ (since $\nabla^2 f = 2I \succeq 2I$). The feasible set $P = \{x : Ax \leq b\}$ is a closed convex set. If P is non-empty, then by Theorem 68: since f is strongly convex, lower semi-continuous, and the domain is closed, the global minimizer exists and is unique (strong convexity gives both existence via coercivity and uniqueness via strict convexity).
(c) We have $\nabla f(x) = 2(x-p)$. By Theorem 69, x^* is optimal iff $\langle \nabla f(x^*), y - x^* \rangle \geq 0$ for all $y \in P$, i.e.:

$$\langle 2(x^* - p), y - x^* \rangle \geq 0 \iff \langle x^* - p, y - x^* \rangle \geq 0.$$

Geometric interpretation: The vector from p to x^* makes an acute angle (or right angle) with every feasible direction $y - x^*$. Equivalently, the hyperplane through x^* with normal $x^* - p$ separates p from the feasible set. This is the projection characterization: x^* is the projection of p onto the convex set P.
(d) Rewrite as $\min \|x\|^2$ s.t. $g_1(x) = -x_1 - x_2 + 4 \leq 0$, $g_2(x) = -2x_1 - x_2 + 5 \leq 0$.
Lagrangian: $\mathcal{L} = x_1^2 + x_2^2 + \lambda_1(-x_1 - x_2 + 4) + \lambda_2(-2x_1 - x_2 + 5)$.
KKT stationarity:

$$2x_1 - \lambda_1 - 2\lambda_2 = 0, \qquad 2x_2 - \lambda_1 - \lambda_2 = 0.$$

Case 1: Both active ($\lambda_1, \lambda_2 > 0$): $x_1 + x_2 = 4$ and $2x_1 + x_2 = 5 \Rightarrow x_1 = 1, x_2 = 3$. From stationarity: $2 = \lambda_1 + 2\lambda_2$ and $6 = \lambda_1 + \lambda_2$. Solving: $\lambda_2 = -4 < 0$. Violates dual feasibility. Rejected.
Case 2: Only g_1 active ($\lambda_2 = 0$): $x_1 + x_2 = 4$. Stationarity: $2x_1 = \lambda_1$, $2x_2 = \lambda_1$, so $x_1 = x_2$. Then $2x_1 = 4 \Rightarrow x_1 = x_2 = 2$, $\lambda_1 = 4 > 0$. Check g_2: $-4 - 2 + 5 = -1 \leq 0$. ✓
Case 3: Only g_2 active ($\lambda_1 = 0$): $2x_1 + x_2 = 5$. Stationarity: $2x_1 = 2\lambda_2$, $2x_2 = \lambda_2$, so $x_1 = 2x_2$. Then $4x_2 + x_2 = 5 \Rightarrow x_2 = 1, x_1 = 2$, $\lambda_2 = 2 > 0$. Check g_1: $-2 - 1 + 4 = 1 > 0$. Infeasible! Rejected.
The closest point is $x^* = (2, 2)^T$ with $\|x^*\|^2 = 8$.

Exercise 5.2 [Hard]
In training a neural network, we minimize a loss $\ell(\theta)$ over parameters $\theta \in \mathbb{R}^D$ subject to a norm constraint to prevent overfitting:

$$\min_{\theta \in \mathbb{R}^D} \ \ell(\theta) \quad \text{s.t.} \quad \|\theta\|_2^2 \leq C.$$

(a) Suppose ℓ is convex. Is this a convex optimization problem? Which problem class (LP, QP, QCQP, SOCP, general convex) does it belong to? Justify.

(b) Suppose furthermore that ℓ is strictly convex. Using the results on convex optimization, what can you conclude about the minimizer?

(c) In practice, neural network losses are *non-convex*. Explain why the theorem "local min $\Rightarrow$ global min" does *not* apply, and give a simple one-dimensional example of a non-convex function with a local minimum that is not global.

(d) Despite non-convexity, practitioners use gradient descent and often find good solutions. Suggest one reason, from an optimization perspective, why this might work. (1–2 sentences suffice.)

Solution

(a) Yes, it is convex. The objective $\ell(\theta)$ is convex by assumption. The constraint $\|\theta\|_2^2 - C \leq 0$ has constraint function $g(\theta) = \theta^T I\theta - C$, which is convex (quadratic with PSD matrix I). No equality constraints. All conditions for a convex optimization problem in standard form are met.
Since ℓ is a general convex function (not necessarily quadratic) with a convex quadratic constraint, this is at least a **QCQP** if ℓ is quadratic, and in general it is a **general convex program**.
(b) If ℓ is strictly convex, then by Theorem 68, if a global minimizer exists, it is unique. Existence is guaranteed because the feasible set $\{\theta : \|\theta\|^2 \leq C\}$ is compact (closed and bounded), and a continuous function on a compact set attains its minimum. Therefore there exists a unique global minimizer.
(c) The theorem requires *both* a convex objective function and a convex feasible set. When ℓ is non-convex, the key step in the proof fails: from $f(y) < f(x^*)$, convexity is used to conclude $f(\lambda y + (1-\lambda)x^*) < f(x^*)$, but this inequality does not hold for non-convex f.
Example: $f(x) = x^4 - 8x^2 + 12$ has a local minimum near $x = 0$ (since $f'(0) = 0$ and we can verify) — actually a cleaner example: $f(x) = (x-1)^2(x+1)^2 - x = x^4 - 2x^2 + 1 - x$.
Simpler: $f(x) = \sin(x) + 0.1x^2$ has multiple local minima, only one of which is global.
(d) One reason is the **over-parameterization** of neural networks: with many more parameters than data points, most local minima tend to have loss values close to the global minimum. Additionally, saddle points (rather than poor local minima) are more common in high dimensions, and gradient descent with stochastic noise tends to escape saddle points.